SUPPLEMENTARY MATERIALS DESCRIPTION

Folder Name-Materials			
S No.	**Description**		
1	Nano indentation data	**Materials** → Nano-indentation test Data → Analysis Sample-pos1 Sample-pos2 Sample-pos3	
2	Experimental data of response	**Materials** → Experimental Data → Damaged_type1 → (16 Data file) Damaged_type2 → (16 Data file) Damaged_type3 → (16 Data file) Undamaged_section → (16 Data file) Comparision_Experimetal_frequencies	
3	Numerical model	**ABAQUS**	**Materials** → ABAQUS model → Damaged_type1 Damaged_type2 Damaged_type3 Undamaged
4		**MATLAB**	**Materials** → MATLAB model → Damaged_type1.xlsx Damaged_type2.xlsx Undamaged.xlsx
5	Frequencies comparison	**Materials** → Frequencies comparison → comparision.xlsx	
6	Dynamic analysis (Acceleration time history comparison	**Materials** → Dynamic analysis → comparision.xlsx Acceleration_comparision.m	
7	Noise data	**Materials** → Noise → Main.m Test_cov2.xlsx	
8	Nonlinear stress-strain data	**Materials** → Nonlinear stress-strain data → ramburg osgoods.xlsx ramburg_osgoods.m	
9	Figures	**Materials** → Figures used	
10	Nodes and sensors Location	**Materials** → Nodal and sensor coordinate	

Data contains following folders as-

- ***Experimental Data:*** *A undamaged_section folder which contains Data file name as und_position1_28th_jan.xlsx with the abbreviations should be read as: und- undamaged, position1 –force is applied at 1st position, 28th_jan – 28th January 2022, similarly for damage types (Folder named as e.g., Damaged_type1) are represented as heavy, lightbeam, reduc for type 1, 2 and 3 damages respectively. Supporting MATLAB codes are also attached in respective folders.*
- ***MATLAB Model:*** *This folder contains 3 excel file as Damaged_type1, Damaged_type2, Undamaged, which represent heavy, light beam and undamaged beam respectively. These folders contain MATLAB files of 3D frame structure.*
- ***ABAQUS Model:*** *This folder contains 3 folders named as Damaged_type1, Damaged_type2, Undamaged, which represent heavy, light beam and undamaged beam respectively. The folder contains ABAQUS files (e.g., project1.cae, project1.jnl) of 3D frame structure.*
- ***Dynamic Analysis:*** *This folder contains Abaqus1.xlsx file in which the acceleration response of Abaqus as well as experimental stored and in MATLAB file (Acceleration_comparision.m) Both responses compared.*
- ***Frequencies comparison:*** *In this folder there is a file named comparision.xlsx, which is created for the comparison of frequencies of damaged and undamaged section.*
- ***Nano-indentation test Data:*** *This data is used for the determination of modulus of elasticity of material. In this folder there is three files named as sample_pos (1,2,3) which means data collected from 3 different position of sample.*
- ***Noise Data:*** *In this folder, there is a file (named main.m) used for the calculation of noise and covariance of noise of the instrument (accelerometer, strain sensor, impact hammer) and a Test_cov.xlsx is the data file of the noise of respective instrument.*
- ***Nonlinear Stress-Strain data:*** *This folder contains 2 file (rambur_osgoods.m and ramburg osgoods.xlsx) which contains actual stress-strain data and optimized ramburg osgoods data of the material.*